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## CE 352P: Transportation Engineering Laboratory

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| Course Code   | CE 352P                                                                                 |
| Course Name   | Transportation Engineering Laboratory                                                   |
| L-T-P-C       | 0-2-0-1                                                                                 |
| Prerequisites | None                                                                                    |
| Intended for  | UG Civil Engineering                                                                    |
| Approval      | 37th Senate (Oct 2022), in the B.Tech Civil Engineering curriculum; 52nd BoA (Nov 2023) |
| Revised       | 57th BoA (Oct 2024) - syllabus revised                                                  |

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### Course Contents

1. Experiments Abrasion Test (Los Angeles, Devel abrasion, etc.)
2. California Bearing Ratio Test for Subgrade (including proctor density)
3. Tests on Bitumen: Penetration
4. Tests on Bitumen: Softening Point
5. Tests on Bitumen: Ductility
6. Tests on Bitumen: Viscosity
7. Tests on Bitumen: Flash and Fire point
8. Tests on Bitumen: Specific Gravity and Water Content
9. Marshal Stability Tests (including rotary evaporator)
10. Banklmen's Beam Test and MERLIN Test
11. Road Survey- Cross section, Super elevation camber, gradient
12. Measuring spot speed/ Flow/ Density/ vehicle count

### Text Books

1. S.K. Khanna, C.E.G. Justo and A. Veeraraghavan, **Highway Engineering**, 10th Edition, Nem Chand Bros., 2018.
2. Kadiyali L.R., **Principles and Practice of highway Engineering**, Khanna Publishers, Delhi, 1992.

### References

1. IS Standards for each test
2. Paul H. Wright and Norman J. Ashford, **Transportation Engineering: Planning and Design**.
3. Nicholas J. Garber and Lester A. Hoel, **Traffic and Highway Engineering**